

Name:

Solutions

Date: _____



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Year 10 Mathematics**Investigation 1, 2015****Topic – Exact Values****Take-Home Section****Important Information:**

Although the complete take-home component is not worth any marks, it is essential in preparation for the in-class validation component. Knowledge and skills gained will be extended in the validation component. This in-class validation will be completed under test conditions on the day in which this take-home component is due. The take home component may be used when completing the validation.

Date out: Week _____ Date _____ **Date Due:** Week _____ Date _____

Take home component weighting: 0% of the year
In-class component weighting: 10 % of the year

In this investigation you will learn how to write answers, that would normally be rounded, in an exact form and why this might be important to do. You will also investigate using and simplify square roots.

Exact Numbers.

An exact number is a number that you can write down without rounding or truncating. It may be a fraction, contain mathematical symbols such as π or $\sqrt{\quad}$ e.g a square root.

Whole numbers and terminating decimals such as 0.5 and 0.25 are exact numbers that we are familiar with. However recurring decimals like 0.3333..., 0.166666..., 0.142857 are examples of numbers that we cannot write down exactly, instead we use symbols to let others know that the number keeps recurring forever. Writing these numbers as decimals gives us an **approximate** answer. The only way to write the recurring decimals as an exact number is if we leave them as fraction. The exact values of the three recurring decimals above are $\frac{1}{3}$, $\frac{1}{6}$ and $\frac{1}{7}$. When asked for an **exact** answer, it is usually best to express your solution as a fraction instead of a decimal.

Exercise 1

a. With the following numbers, classify them as either **exact** or **approximate**.

	Number	Exact/Approximate		Number	Exact/Approximate
i	1.5	E	ii	$\frac{1}{7}$	E
iii	$1.\bar{5}$	A	iv	$14.\bar{2}$	A
v	$\sqrt{7}$	E	vi	3.1414...	A

- b. Find 5 terminating decimals and 5 recurring decimals. Give your answers as both a decimal and a fraction.

Terminating Decimals		Recurring Decimals	
Decimal	Fraction	Decimal	Fraction

- c. Calculate the following. Leave your solution as an **exact** value.

i	$240 \div 4$ 60	ii	$8 \div 32$ $\frac{1}{4}$ or 0.25
iii	$3 \div 48$ 0.0625 or $\frac{1}{16}$	iv	$5 \div (20 \times 100\,000)$ 0.0000025 or $\frac{1}{400\,000}$
v	$3 \times 2 \div 7$ $\frac{6}{7}$	vi	$22 \div (11 \times 3)$ $\frac{2}{3}$
vii	$5 \div 13 \times 2$ $\frac{5}{26}$	viii	$\frac{4-3}{12-5}$ $\frac{1}{7}$
ix	$\sqrt{121}$ 11	x	$\sqrt{25-9}$ 4

Exact values with square roots

Most square roots will not give you an exact answer. The two in exercise 2b did, only because the numbers 121 and 16 are square numbers, however if I put $\sqrt{2}$ into my calculator I get 1.414213562. This number has been rounded because my calculator can only display 10 digits, and thus is not an exact number. You will have the similar issue with $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$ and $\sqrt{7}$, these numbers belong to a set of numbers called surds. They are numbers that produce decimals that do not recur and do not terminate.

Exercise 2

In the table below find 4 square roots, that have not appeared in this investigation that give exact answers when evaluated with a calculator and also find 4 more square roots that don't give exact answers when evaluated with a calculator (surds).

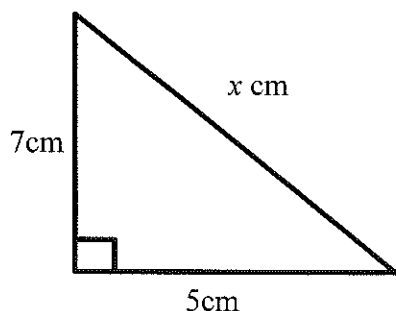
Square roots with exact answers		Square roots without exact answers	
Square root	Calculated answer	Square root	Calculated answer
Numbers will be a square numbers		Numbers will not be square numbers.	

Why do we need exact values?

- If we need to use a rounded number in a calculation, we will get the wrong answer. Sometimes the rounding of the number is enough to get the whole question wrong.
- Exact numbers are usually easier to write down and easier to enter into calculators.

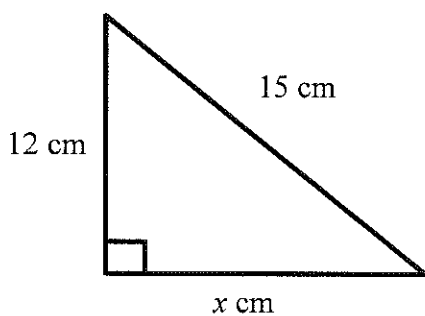
So when a question asks for an exact answer, we can leave it as a fraction or a square root.

Example



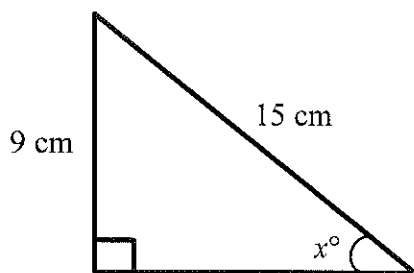
Find the **exact** value of x

$$\begin{aligned}
 c^2 &= a^2 + b^2 \\
 x^2 &= 7^2 + 5^2 \\
 &= 49 + 25 \\
 &= 74 \\
 x &= \sqrt{74} \text{ cm}
 \end{aligned}$$



Find the **exact** value of x

$$\begin{aligned} a^2 &= c^2 - b^2 \\ x^2 &= 15^2 - 12^2 \\ &= 225 - 144 \\ &= 81 \\ x &= \sqrt{81} \\ &= 9 \text{ cm} \end{aligned}$$



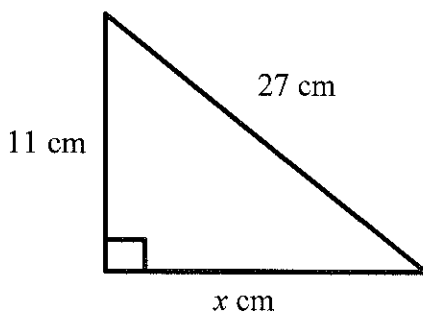
State the value of $\sin x$ as a ratio in simplest form

$$\begin{aligned} \sin \theta &= \frac{\text{opp}}{\text{hyp}} \\ \sin x &= \frac{9}{15} \\ &= \frac{3}{5} \end{aligned}$$

Exercise 3

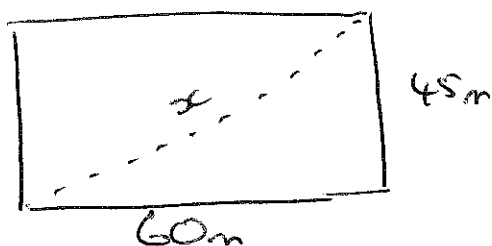
Solve the following, leave all answers as **exact** values.

- a. Find the exact value of x



$$\begin{aligned} x^2 &= 27^2 - 11^2 \\ &= 608 \\ x &= \sqrt{608} \\ &= 4\sqrt{38} \quad (\text{if calculator simplifies}) \end{aligned}$$

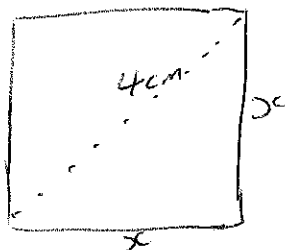
- b. Calculate the exact value of the diagonal across a rectangular paddock that has a width of 45 meters and a length of 60 meters (Hint: Draw a diagram)



$$\begin{aligned} x^2 &= 60^2 + 45^2 \\ &= 5625 \\ x &= \sqrt{5625} \\ &= 75 \text{ m} \end{aligned}$$

c. The side of a cube has a diagonal length of 4cm

- i. Use Pythagoras to calculate the **exact** value of the side of the cube
(Hint draw a square, label the sides x and the diagonal 4)



$$\begin{aligned} 4^2 &= x^2 + x^2 \\ 16 &= x^2 + x^2 \\ \therefore x^2 &= 8 \\ x &= \sqrt{8} \end{aligned}$$

Exact length = $\sqrt{8}$ cm

- ii. Round your answer in i. above to 1 decimal place.

Approximate length = 2.8 cm

- iii. Using your exact answer from i above, calculate the exact volume of the cube
Hint: find x^3 and $(\sqrt{a})^n = \sqrt{a^n}$

$$\begin{aligned} x^3 &= (\sqrt{8})^3 \\ &= \sqrt{8^3} \\ &= \sqrt{512} \\ &= 16\sqrt{2} \quad (\text{if calculator simplifies}) \end{aligned}$$

- iv. What is the volume of the cube to the nearest cm?

$$\sqrt{512} = 22.627 \text{ (sdp)} \therefore 23 \text{ cm}^3 \text{ to nearest cm}$$

- v. Calculate the volume using the approximate answer in ii.

$$\begin{aligned} 2.8^3 &= 21.952 \text{ (sdp)} \\ &= 22 \text{ cm}^3 \text{ to nearest cm} \end{aligned}$$

- vi. Does the rounded answer in v. give the same value as the exact solution in iv?

No approximate gives wrong answer

Simplifying expressions involving square roots

Now that we have solutions with square roots in them, sometimes we need to add the answer of one problem to the answer of another. To do this without losing our accuracy and keeping our exact answers we need to fall back on our algebra skills.

Since we started learning algebra we have add like terms together, as follows.

$$3x + 5x = 8x$$

$$2y + 7y = 9y$$

$$10r - 3r = 7r$$

The same process can be applied to expressions involving square roots. We treat the $\sqrt{\quad}$ as the letter.

$$3\sqrt{2} + 5\sqrt{2} = 7\sqrt{2}$$

$$2\sqrt{3} + 7\sqrt{3} = 9\sqrt{3}$$

$$10\sqrt{11} - 3\sqrt{11} = 7\sqrt{11}$$

So the $\sqrt{2}$ is treated like the x , the $\sqrt{3}$ is treated like the y , the $\sqrt{11}$ is treated like the r .

Exercise 4

a. $5\sqrt{2} + 7\sqrt{2}$

$$12\sqrt{2}$$

b. $12\sqrt{17} - 3\sqrt{17}$

$$9\sqrt{17}$$

c. $16\sqrt{7} + \sqrt{7}$

$$17\sqrt{7}$$

d. $35\sqrt{23} - 9\sqrt{23}$

$$26\sqrt{23}$$

e. $54\sqrt{13} + 6\sqrt{13} + 11\sqrt{13}$

$$71\sqrt{13}$$

f. $15\sqrt{31} - 19\sqrt{31} + 8\sqrt{31}$

$$4\sqrt{31}$$

g. $7\sqrt{3} + 5\sqrt{3} - 2\sqrt{3} + 9\sqrt{3}$

$$19\sqrt{3}$$

Just as in Algebra, where you can only add terms with the same letters, you can **only** add square roots with the **same number** under the root sign.

Algebra Examples:

1. $3x + 2y = 3x + 2y$

(cannot simplify further)

2. $4x + 7y - x + 2y = 4x - x + 7y + 2y$
 $= 3x + 9y$

(rearrange so like terms are together)
 (cannot simplify further)

Square Root Examples:

1. $3\sqrt{2} + 2\sqrt{5} = 3\sqrt{2} + 2\sqrt{5}$

(cannot simplify further)

2. $4\sqrt{2} + 7\sqrt{5} - \sqrt{2} + 2\sqrt{5} = 4\sqrt{2} - \sqrt{2} + 7\sqrt{5} + 2\sqrt{5}$
 $= 3\sqrt{2} + 9\sqrt{5}$

(rearrange like terms)
 (cannot simplify further)

Exercise 5

a. $12\sqrt{2} + 15\sqrt{7} - 4\sqrt{2} + 9\sqrt{7}$
 $12\sqrt{2} - 4\sqrt{2} + 15\sqrt{7} + 9\sqrt{7}$
 $8\sqrt{2} + 24\sqrt{7}$

b. $13\sqrt{6} - 6\sqrt{10} + \sqrt{6} - 3\sqrt{10}$
 $13\sqrt{6} + \sqrt{6} - 6\sqrt{10} - 3\sqrt{10}$
 $14\sqrt{6} - 9\sqrt{10}$

c. $\sqrt{17} + 5\sqrt{3} + 2\sqrt{17} + 14\sqrt{3}$
 $\sqrt{17} + 2\sqrt{17} + 5\sqrt{3} + 14\sqrt{3}$
 $3\sqrt{17} + 19\sqrt{3}$

d. $9\sqrt{7} + 6\sqrt{8} - 7\sqrt{7} - 7\sqrt{8}$
 $9\sqrt{7} - 7\sqrt{7} + 6\sqrt{8} - 7\sqrt{8}$
 $2\sqrt{7} - \sqrt{8}$

e. $21\sqrt{12} + 32\sqrt{5} - 3\sqrt{12} + 2\sqrt{5}$
 $21\sqrt{12} - 3\sqrt{12} + 32\sqrt{5} + 2\sqrt{5}$
 $18\sqrt{12} + 34\sqrt{5}$

f. $12\sqrt{14} - 5\sqrt{11} - 8\sqrt{14} - \sqrt{11}$
 $12\sqrt{14} - 8\sqrt{14} - 5\sqrt{11} - \sqrt{11}$
 $4\sqrt{14} - 6\sqrt{11}$

Simplifying Square Roots

Recall from the rules of indices that: $(a \times b)^n = a^n \times b^n$

$$\sqrt{a} = a^{\frac{1}{2}} \text{ and } \sqrt{b} = b^{\frac{1}{2}}$$

So we can write:

$$\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$$

This is very useful for simplifying square roots. In other words, we can split a single square root into the product of the roots of its factors.

Example

$$\sqrt{20} = \sqrt{4 \times 5} = \sqrt{4} \times \sqrt{5}$$

$$\sqrt{18} = \sqrt{6 \times 3} = \sqrt{6} \times \sqrt{3}$$

$$\sqrt{40} = \sqrt{4 \times 10} = \sqrt{4} \times \sqrt{10}$$

Exercise 6

Break the following square roots into the product of two square roots of its factors. Check on your calculator that both the question and your answer give the same decimal value.

a.	$\sqrt{6}$ $\sqrt{2} \times \sqrt{3}$	b.	$\sqrt{26}$ $\sqrt{2} \times \sqrt{13}$
c.	$\sqrt{55}$ $\sqrt{5} \times \sqrt{11}$	d.	$\sqrt{21}$ $\sqrt{3} \times \sqrt{7}$
e.	$\sqrt{38}$ $\sqrt{2} \times \sqrt{19}$	f.	$\sqrt{52}$ $\sqrt{26} \times \sqrt{2}$ or $\sqrt{13} \times \sqrt{4}$
g.	$\sqrt{93}$ $\sqrt{31} \times \sqrt{3}$	h.	$\sqrt{108}$ $\sqrt{54} \times \sqrt{2}$ or $\sqrt{27} \times \sqrt{4}$

However, we have not simplified the square root, merely created two of them! For this process to be useful, one of the factors needs to be a square number.

It may be useful at this point to list the first 12 square numbers. They are:

Square Number's	Calculation
1	1×1
4	2×2
9	3×3
16	4×4
25	5×5
36	6×6
49	7×7
64	8×8
81	9×9
100	10×10
121	11×11
144	12×12

With the exception of 1, the square numbers are the numbers we want to find for one of the factors of our original number.

Example: $20 = 4 \times 5$ ↖ square number

$$27 = 9 \times 3$$

$$8 = 4 \times 2$$

Exercise 7			
Factorise the following so that one factor is a square number			
a.	18		9×2
b.	40		4×10
c.	32		16×2
d.	98		49×2
e.	75		25×3
f.	288		144×2

We now have all the skills to **simplify** square roots.

As we did earlier $\sqrt{20}$ can be written as $\sqrt{4 \times 5}$, a square number and another number. This can be re-written as $\sqrt{4} \times \sqrt{5}$. Since we can evaluate $\sqrt{4}$, we end up with $2 \times \sqrt{5}$. Finally we remove the multiplication sign which leaves $2\sqrt{5}$ as our final simplified answer.

Examples: **Simplify $\sqrt{40}$**

$$\begin{aligned}\sqrt{40} &= \sqrt{4 \times 10} \\ &= \sqrt{4} \times \sqrt{10} \\ &= 2 \times \sqrt{10} \\ &= 2\sqrt{10}\end{aligned}$$

Simplify $\sqrt{18}$

$$\begin{aligned}\sqrt{18} &= \sqrt{9 \times 2} \\ &= \sqrt{9} \times \sqrt{2} \\ &= 3 \times \sqrt{2} \\ &= 3\sqrt{2}\end{aligned}$$

Exercise 8			
Simplify the following square roots			
a.	$\begin{aligned}\sqrt{8} &= \sqrt{4 \times 2} \\ &= \sqrt{4} \times \sqrt{2} \\ &= 2\sqrt{2}\end{aligned}$	b.	$\begin{aligned}\sqrt{12} &= \sqrt{4 \times 3} \\ &= \sqrt{4} \times \sqrt{3} \\ &= 2\sqrt{3}\end{aligned}$
c.	$\begin{aligned}\sqrt{27} &= \sqrt{9 \times 3} \\ &= \sqrt{9} \times \sqrt{3} \\ &= 3\sqrt{3}\end{aligned}$	d.	$\begin{aligned}\sqrt{75} &= \sqrt{25 \times 3} \\ &= \sqrt{25} \times \sqrt{3} \\ &= 5\sqrt{3}\end{aligned}$
e.	$\begin{aligned}\sqrt{128} &= \sqrt{64 \times 2} \\ &= \sqrt{64} \times \sqrt{2} \\ &= 8\sqrt{2}\end{aligned}$	f.	$\begin{aligned}\sqrt{363} &= \sqrt{121 \times 3} \\ &= \sqrt{121} \times \sqrt{3} \\ &= 11\sqrt{3}\end{aligned}$